

ECON100B - SECTION 10

The Romer Model Wrap Up and PSET/Exam Practice

Felix Zhang

ANNOUNCEMENTS/AGENDA

Announcements

- PSET 2 due Friday, 2/27
- In-Class Midterm, 3/5

Agenda

- Romer Model PSET problem
- Practice and PSET

Question of the Day

Under the Romer model, if we increased the parameter $\bar{l}$, the percentage of people working in the production of knowledge and ideas, what happens to output in the short run? What about the long run? What happens to the growth rate of ideas?

If you could become any animal for a day, which animal would it be?

The Romer Model Wrap Up



PSET 2, Q3

3. [10 points total] The Romer Model and the Zombie Apocalypse.

Braaaaaaaains! Consider an “old school” Romer economy² like the one we saw in class, with production of knowledge A and food Y . Scientists produce new knowledge using old knowledge and have productivity $\bar{z}$. Total population is fixed at $\bar{N}$ and employed either in knowledge or in food. The share of population in knowledge production equals ℓ , a number like 0.10.

$$\Delta A_t = \bar{z} A_t L_{at} \quad (3.1)$$

$$Y_t = A_t L_{yt} \quad (3.2)$$

$$\bar{N} = L_{at} + L_{yt} \quad (3.3)$$

$$L_{at} = \ell \bar{N} \quad (3.4)$$

Recall that the solution to the Romer Model reveals the following:

$$g[A] = \bar{z} \ell \bar{N} \quad (3.5)$$

$$A_t = A_0 (1 + g[A])^t \quad (3.6)$$

$$y_t = A_0 (1 + g[A])^t (1 - \ell) \quad (3.7)$$

Imagine what might happen if all the scientists accidentally infect themselves with the “zombie flu” and become zombies. Zombies:

- Do not produce new ideas
- Do not destroy old ideas
- Eat brains and do not eat food produced by farmers
- Do not eat farmers’ brains. They prefer tasty imported brains

PSET 2, Q3

- (a) [3 points] When the scientists become zombies, what happens to the growth rate of A ? What about the growth rate of y ?
- (b) [3 points] When the scientists become zombies, what happens to the level of A ? What about the level of output per person? (Consult the bullet points about zombies above.)

PSET 2, Q3

- (c) [4 points] Illustrate the Zombie Apocalypse in the graph below, showing any changes in the levels and growth rates in output per person over time. Discuss what you have drawn. Is the zombie flu good or bad? Explain.

PSET 2, Q3

(d) [4 points] Suppose rather than being turned into zombies, the scientists had all been deported to another country by the government. Would anything be different? Why?

(e) [4 points] Suppose rather than being turned into zombies, the scientists had all been repurposed as farmers by the government. Would anything be different? Why?

PSET 2, Q2

- (a) [4 points] On the axes below, draw the Solow diagram identifying the saving, depreciation, and income curves, and the steady state. Briefly explain what you have drawn.

PSET 2, Q2

- (b) [6 points] Now redraw the Solow diagram showing the effect(s), if any, of a one-time increase in the labor supply, L , due to a substantial amount of immigration. Assume nothing else changes, and explain, discussing short and long-run effects.

PSET 2, Q2

So far, we have assumed that immigrants save at the same rate as natives, which seems reasonable. But what would happen if immigrants sent some of their saving back to their native countries, where it became foreign capital?

- (c) [4 points] Redraw part (b) showing the full effects of immigration: the increase in L and any change in the national investment rate s owing to immigration. On the next page, explain what you have drawn, discussing short and long-run effects.

PSET 2, Q2

(d) [2 points] In the steady state, what is true about ΔK ? Rewrite equation (2.2) using this.

(e) [2 points] Substitute equation (2.1) into your answer to part (d) to find an equation that includes A , K , L , s , and d .

PSET 2, Q2

- (f) [2 points] Manipulate your answer to part (e) to find an equation for the steady-state level of capital, K^* , in terms of A , L , s , and d . Show your work.

PSET 2, Q2

- (g) [2 points] Take the growth rate of both sides of your answer to part (f) to find a formula for the growth rate of K in terms of the growth rates of A , L , s , and d . Simplify.
-
- (h) [2 points] Take the growth rate of both sides of equation (1) to find a formula for the growth rate of output, Y , in terms of the growth rates of A , K , and L . Simplify.
- (i) [2 points] (i) (2 points) Finally, subtract the growth rate of L from your answer to part (h) to find a formula for the growth rate of output per person, Y/L , in terms of the growth rates of A , K , and L . Simplify.

PSET 2, Q2

Today, the [share of Americans who are foreign-born is around 15-16 percent](#). For simplicity, let us assume that statistic is 10 percent, and although immigrants actually arrive in much smaller numbers, let us examine the effects of a one-time increase in L equal to 10 percent.

- (j) [3 points] Ignoring any change in the national saving rate, what happens to output per person in the short run when L rises by 10 percent? Use your answer to part (i) and be specific.

- (k) [3 points] Ignoring any change in the national saving rate, what happens to output per person in the long run when L rises by 10 percent? Use your answers to parts (g) and (i) and be specific.

PSET 2, Q2

Now let us think about potential effects of immigration on saving and capital. Assume the original saving rate s among U.S. natives was 20 percent, and let us take an extreme view: that all immigrants send all of their saving abroad. Thus their contribution to the national investment rate is zero.

- (l) [3 points] If 90 percent of residents save at 20 percent, and 10 percent of residents save at 0 percent, what is the new national or average saving rate, or is it the same? (Hint: this is like reducing the number of savers by the percent who are immigrants.)

- (m) [5 points] What are the long-run effects of immigration on output per person if the saving rate does as you have described in part (l)? Use your answers to parts (g) and (i) and be specific.

2012 Midterm Practice

3. (30 points total) *Korean Reunification in the Solow Model.*

North and South Korea have been separate nations roughly since the end of World War II, but some analysts believe reunification, such as occurred between the former East and West Germany in 1990, is a likely prospect sometime in the future. What are the macroeconomic implications of such a development?

Take the perspective of South Korea, a country relatively rich in capital (K). Recall that the key equations of the Solow Model are:

$$Y = A K^{1/3} L^{2/3} \quad (2)$$

$$\Delta K = sY - dK \quad (3)$$

- (a) (4 points) On the axes below, draw the Solow diagram for South Korea *prior to reunification*, identifying the saving curve, depreciation curve, income curve, and the steady state.

2012 Midterm Practice

North Korea's population is about 25 million people, or about half the level of population in South Korea. But the GDP of North Korea is only about one quarter the size of the South's.

- (b) (3 points) What does this reveal about GDP per person in the North versus the South? Be specific.

- (c) (3 points) State two possible explanations for the result in part (b) based on the production function in equation (2) and briefly explain.

2012 Midterm Practice

Peaceful reunification! Suppose that overnight, North Korean authorities, like their East German counterparts did many years ago, stop enforcing the border, and imagine that the entire population of North Korea picks up and moves itself to South Korea overnight, bringing nothing with them but the clothes on their backs and optimism for the future.

- (d) (4 points) On the axes below, draw a new Solow diagram for South Korea *showing the effects of reunification*, identifying the saving curve(s), depreciation curve(s), income curve(s), and the steady state(s).

2012 Midterm Practice

- (e) (3 points) Using equation (2), derive an equation for output per worker as a function of productivity (A) and capital per worker. Show your work.

2012 Midterm Practice

(f) (3 points) *Qualitatively speaking*, what happens *in the short run*, immediately after the arrival of North Koreans, to (1) the wage and (2) income per capita in South Korea, if anything? Why?

(g) (3 points) Take growth rates of both sides of your answer to part (e) and equation (4) in part (f) to find *numerical answers* for the effects of reunification on (1) the wage and (2) income per capita. Recall that North Korea's population is 50% the size of South Korea's.

2012 Midterm Practice

Although politically divided for more than 60 years, North and South Koreans share a common cultural ancestry and arguably have similar preferences and tastes.

- (h) (3 points) What do you think will be true about the *saving rates* of former North Koreans compared to original South Koreans now that all are in the South? Explain.

- (i) (4 points) Based on your answer to part (h), what happens to the wage and per capita income in South Korea *in the long run*? (You get full credit only if you use the mathematical definition of the steady state in your argument.)